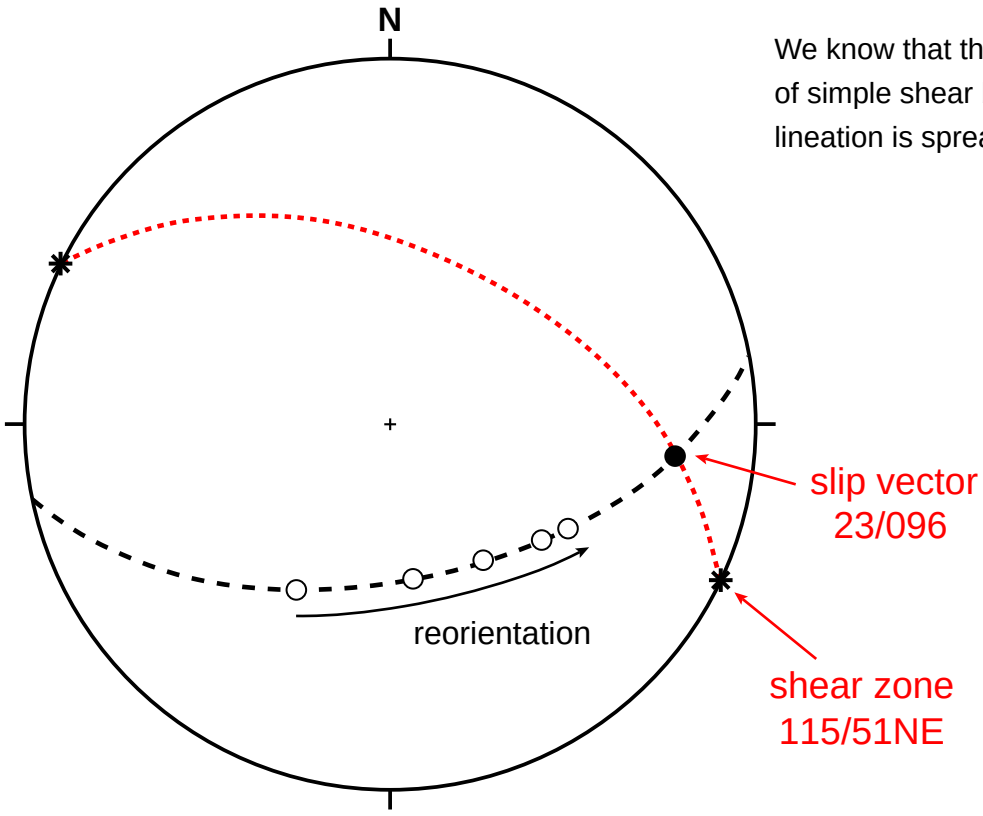
SHEAR ZONE EXERCISE

The outcrop trace of a plastic shear zone is 115° and a mineral lineation developed on mylonites within it plunges 23/096. The following measurements of a lineation that has been reoriented by the shearing were made on a traverse starting from outside the shear zone and walking towards the zone: 47/209, 55/171, 53/145, 46/127, 43/120

Determine (a) the orientation of the shear zone, and (b) the sense of movement within it. How do we know that the shear zone is truly a zone of simple shear?

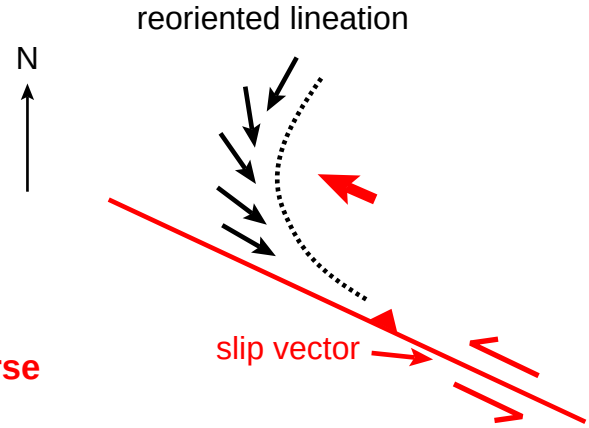


We know that the shear zone is truly a zone of simple shear because the reoriented lineation is spread along a great circle

✱ strike of shear zone

● mineral lineation on mylonite = slip vector

○ reoriented lineation



Shear sense is sinistral and reverse